

AVERAGE, MIXTURE & ALLIGATION

Fully Solved Questions with Step-by-Step Explanation & Shortcut Tricks

How to use these notes: Every question is solved in full steps first, followed by a ✨ Trick box that gives the fastest exam method for that concept. Study the trick after understanding the full solution.

SECTION A — AVERAGE

Q1. Vibhor obtained 66, 74, 55, 92, 79 (out of 100) in English, Maths, Physics, Chemistry and Biology. What's his average marks?

(a) 75% (b) 69% (c) 73.2% (d) 76%

Average = Sum of all observations ÷ Number of observations

$$\text{Sum} = 66 + 74 + 55 + 92 + 79 = 366$$

$$\text{Average} = 366 \div 5 = 73.2$$

Since each subject is out of 100, the average marks = 73.2%

Answer: (c) 73.2%

⚡ **Trick / Shortcut**

Whenever marks are 'out of 100', average marks and average percentage are the same number — no separate percentage calculation needed.

Add numbers in pairs to avoid mistakes: $(66+74)=140$, $(55+92)=147$, $140+147+79=366$.

Q2. Find the average of all the numbers between 8 and 55 which are divisible by 7.

(a) 17 (b) 31.5 (c) 34 (d) 44

Numbers between 8 and 55 divisible by 7: 14, 21, 28, 35, 42, 49

These numbers form an Arithmetic Progression (AP) with common difference 7.

For any AP, Average = (First term + Last term) ÷ 2

$$\text{Average} = (14 + 49) \div 2 = 63 \div 2 = 31.5$$

Answer: (b) 31.5

⚡ **Trick / Shortcut**

GOLDEN RULE: For evenly spaced numbers (consecutive numbers, consecutive multiples, an AP), NEVER add all terms.

Just use: Average = (First term + Last term) ÷ 2. This saves a lot of time in exams.

Q3. The average of first five multiples of 5 is:

(a) 10 (b) 12 (c) 15 (d) 25

First five multiples of 5: 5, 10, 15, 20, 25

This is an AP — apply the same shortcut.

$$\text{Average} = (\text{First} + \text{Last}) \div 2 = (5 + 25) \div 2 = 30 \div 2 = 15$$

Answer: (c) 15

★ **Trick / Shortcut**

For the first n multiples of any number k , average = $k \times (n+1)/2$.

Here: $5 \times (5+1)/2 = 5 \times 3 = 15$. Instant answer without listing numbers.

Q4. The average of 5 consecutive numbers is 30. The largest of these numbers is:

(a) 20 (b) 22 (c) 32 (d) 40

For an odd count of consecutive numbers, the average is always the middle (3rd) number.

So the middle number = 30, and the numbers are 28, 29, 30, 31, 32.

Largest number = 32

Answer: (c) 32

★ **Trick / Shortcut**

For n consecutive numbers (n odd) with average A : Largest = $A + (n-1)/2$, Smallest = $A - (n-1)/2$.

Here: Largest = $30 + (5-1)/2 = 30 + 2 = 32$.

Q5. The sum of four consecutive odd numbers is 30 more than the average of these numbers. What is the first of these numbers?

(a) 7 (b) 14 (c) 17 (d) 20

Let the four consecutive odd numbers be $x, x+2, x+4, x+6$.

Sum = $4x + 12$ and Average = $(4x+12)/4 = x + 3$

Given: Sum = Average + 30 $\rightarrow 4x + 12 = (x + 3) + 30$

$4x + 12 = x + 33 \rightarrow 3x = 21 \rightarrow x = 7$

First number = 7

Answer: (a) 7

★ **Trick / Shortcut**

Always assume consecutive odd/even numbers as $x, x+2, x+4, x+6 \dots$ (common difference 2) — this keeps algebra simple.

Sum of n consecutive numbers is always $n \times (\text{average})$, so 'Sum - Average' straightaway gives $(n-1) \times \text{average}$.

Q6. The mean of $1^3, 2^3, 3^3, 4^3, 5^3$ is:

(a) 30 (b) 35 (c) 45 (d) 70

Sum of cubes of first n natural numbers = $[n(n+1)/2]^2$

For $n = 5$: Sum = $[5 \times 6/2]^2 = 15^2 = 225$

Mean = $225 \div 5 = 45$

Answer: (c) 45

★ **Trick / Shortcut**

Memorise: $\Sigma n = n(n+1)/2$, $\Sigma n^2 = n(n+1)(2n+1)/6$, $\Sigma n^3 = [n(n+1)/2]^2$.

These formulas convert a long addition into a 10-second calculation — extremely useful for averages of squares/cubes.

Q7. The average age of 14 girls and their teacher's age is 15 years. If the teacher's age is excluded, the average reduces by 1. What is the teacher's age?

(a) 32 years (b) 30 years (c) 29 years (d) 35 years

Total people = 14 girls + 1 teacher = 15

Total age of all 15 = $15 \times 15 = 225$ years

Average of 14 girls only = $15 - 1 = 14$, so total age of 14 girls = $14 \times 14 = 196$ years

Teacher's age = $225 - 196 = 29$ years

Answer: (c) 29 years

✨ **Trick / Shortcut**

MASTER TRICK for all such problems: convert every average into a TOTAL (Average $\times$ Number of people) first.
Then simple addition/subtraction of totals gives the answer — never work with averages directly.

Q8. The average marks obtained by 40 students of a class is 86. If the 5 highest marks are removed, the average reduces by one mark. The average marks of the top 5 students is:

(a) 92 (b) 96 (c) 93 (d) 97

Total marks of 40 students = $40 \times 86 = 3440$

After removing top 5, remaining 35 students have average = $86 - 1 = 85$

Total marks of remaining 35 students = $35 \times 85 = 2975$

Sum of marks of top 5 students = $3440 - 2975 = 465$

Average of top 5 = $465 \div 5 = 93$

Answer: (c) 93

✨ **Trick / Shortcut**

Same Total-based approach: Total(all) - Total(remaining) = Total(removed group).
Always identify: what changed (count) and what changed (average) — then just work with totals.

Q9. A student finds the average of 10, 2-digit numbers. If the digits of one of the numbers are interchanged, the average increases by 3.6. The difference between the digits of that 2-digit number is:

(a) 4 (b) 3 (c) 2 (d) 5

Average increases by 3.6 for 10 numbers $\rightarrow$ total sum increases by $3.6 \times 10 = 36$

Let the original number be $10a + b$ (tens digit a , units digit b). After interchanging: $10b + a$.

Increase = $(10b + a) - (10a + b) = 9(b - a) = 36$

$b - a = 4$, so the difference between the digits is 4

Answer: (a) 4

✨ **Trick / Shortcut**

Whenever digits of a 2-digit number are interchanged, the value always changes by a multiple of 9: $|10a+b - (10b+a)| = 9|a-b|$.
So directly divide the change in sum by 9 to get the digit difference.

Q10. If the average marks of three batches of 40, 50 and 60 students respectively is 30, 50 and 70, then the average marks of all the students is:

(a) 63.33 (b) 52.66 (c) 60 (d) 65

Total marks = $(40 \times 30) + (50 \times 50) + (60 \times 70) = 1200 + 2500 + 4200 = 7900$

Total students = $40 + 50 + 60 = 150$

Combined average = $7900 \div 150 = 52.66$

Answer: (b) 52.66

⚡ **Trick / Shortcut**

Weighted average formula: $(n_1A_1 + n_2A_2 + n_3A_3) / (n_1 + n_2 + n_3)$.

Never simply average the three averages (30, 50, 70 $\rightarrow$ 50) — that ignores the different group sizes and is a common exam trap.

Q11. The average age of A and B is 20 years. If A is to be replaced by C, the average would be 19 years. The average age of C and A is 21 years. The ages of A, B and C in order (in years) are:

(a) 18, 22, 20 (b) 18, 20, 22 (c) 22, 18, 20 (d) 22, 20, 18

$$A + B = 20 \times 2 = 40 \text{ ... (i)}$$

$$B + C = 19 \times 2 = 38 \text{ ... (ii) [A replaced by C]}$$

$$C + A = 21 \times 2 = 42 \text{ ... (iii)}$$

$$\text{Adding (i)+(ii)+(iii): } 2(A+B+C) = 120 \rightarrow A+B+C = 60$$

$$C = 60 - (A+B) = 60 - 40 = 20; A = 60 - (B+C) = 60 - 38 = 22; B = 60 - (C+A) = 60 - 42 = 18$$

$$\text{So } A = 22, B = 18, C = 20$$

Answer: (c) 22, 18, 20

⚡ **Trick / Shortcut**

When three pair-sums are given for A, B, C: add all three equations and divide by 2 to get (A+B+C) directly.

Then each individual value = (A+B+C) - (the pair not containing it). Very fast for these 'replacement' average questions.

Q12. The average weight of 21 boys was recorded as 64 kg. If the weight of the teacher was added, the average increased by one kg. What was the teacher's weight?

(a) 86 Kg (b) 64 Kg (c) 72 Kg (d) 98 Kg

$$\text{Total weight of 21 boys} = 21 \times 64 = 1344 \text{ kg}$$

$$\text{After adding teacher, total people} = 22, \text{ new average} = 64 + 1 = 65 \text{ kg}$$

$$\text{New total weight} = 22 \times 65 = 1430 \text{ kg}$$

$$\text{Teacher's weight} = 1430 - 1344 = 86 \text{ kg}$$

Answer: (a) 86 Kg

⚡ **Trick / Shortcut**

Fast formula when ONE new member is added and average rises by 'd': New member's value = Old average + $n \cdot d$ (n = old number of members after adding, i.e., new count).

Check: $64 + 22 \times 1 = 86$. Matches instantly — no need to compute both totals separately.

Q13. The average age of X and Y is 33 years. If Z's age is added the average of their ages becomes 31 years. What is the age of Z?

(a) 24 years (b) 27 years (c) 30 years (d) 33 years

$$X + Y = 33 \times 2 = 66$$

$$X + Y + Z = 31 \times 3 = 93$$

$$Z = 93 - 66 = 27 \text{ years}$$

Answer: (b) 27 years

⚡ **Trick / Shortcut**

Same 'added member' shortcut: $\text{New value} = \text{New average} + (\text{old count}) \times (\text{change in average, with sign})$.

Here average fell by 2 for 2 old members: $Z = 31 + 2 \times (-2)$... simpler to just subtract totals as shown above.

Q14. The average of 11 results is 50. If the average of the first six results is 49 and that of the last six is 52, the sixth number is:

(a) 48 (b) 50 (c) 52 (d) 56

Total of 11 results = $11 \times 50 = 550$

Total of first 6 results = $6 \times 49 = 294$

Total of last 6 results = $6 \times 52 = 312$

First-6 + Last-6 totals = $294 + 312 = 606$, but the 6th number is counted twice in this sum

6th number = $606 - 550 = 56$

Answer: (d) 56

⚡ **Trick / Shortcut**

Whenever 'first k' and 'last k' groups overlap in one number (out of n total, with $n < 2k-1$), that repeated number = $\text{Total}(\text{first } k) + \text{Total}(\text{last } k) - \text{Total}(\text{all } n)$.

This overlap trick appears very often — memorise it as a standard formula.

Q15. The average of 15 players' points at a football match was 15, that of the 1st 8 being 20 and that of the last 8 being 18. What was the 8th player's observation?

Options: 56, 77, 89, 79

Total of 15 players = $15 \times 15 = 225$

Total of first 8 players = $8 \times 20 = 160$

Total of last 8 players = $8 \times 18 = 144$

Sum of both groups = $160 + 144 = 304$, with the 8th player counted twice (common to both groups)

8th player's score = $304 - 225 = 79$

Answer: 79

⚡ **Trick / Shortcut**

Same overlap formula as Q14: $\text{Repeated term} = \text{Total}(\text{first group}) + \text{Total}(\text{last group}) - \text{Total}(\text{all})$.

Recognising 'first 8 of 15' and 'last 8 of 15' overlapping in exactly the 8th player is the key insight.

Q16. Rishabh Pant made a score of 94 runs in the 20th innings of an Australia vs India Test match and thus increased his average by 2. Find his average after the 20th innings.

Options: 47, 58, 56, 60

Let average after 19 innings = x. Then average after 20 innings = $x + 2$.

Total after 19 innings = $19x$. Total after 20 innings = $19x + 94$.

Also, Total after 20 innings = $20(x + 2)$

So $19x + 94 = 20(x + 2) = 20x + 40$

$94 - 40 = 20x - 19x \rightarrow x = 54$

Average after 20th innings = $x + 2 = 56$

Answer: 56

✦ **Trick / Shortcut**

Direct formula: $\text{New Average} = \text{New Score} - (\text{Increase} \times \text{New number of innings})$

Check: $94 - (2 \times 20) = 94 - 40 = 54$ (old average); new average = $54 + 2 = 56$. This one-line formula avoids setting up equations.

SECTION B — MIXTURE & ALLIGATION

Q17. A mixture of 160 gallons of wine and water contains 25% water. How much water must be added to the mixture to increase the percentage of water to 40% of the new mixture?

Options: 40 gals, 50 gals, 80 gals, 33 gals

Water in original mixture = 25% of 160 = 40 gallons; Wine = 120 gallons (wine quantity never changes).

Let water added = w . New total mixture = $160 + w$; New water quantity = $40 + w$.

Required: $(40 + w) / (160 + w) = 40/100 = 0.4$

$40 + w = 64 + 0.4w \rightarrow 0.6w = 24 \rightarrow w = 40$

Water to be added = 40 gallons

Answer: 40 gals

✦ **Trick / Shortcut**

The quantity of the ingredient that is NOT being added (here, wine) stays constant — always anchor your equation on that constant quantity.

Quick check formula: $\text{New water\%} \times \text{New total} = \text{New water quantity}$. Plug and solve for the unknown added quantity.

Q18. 800 students took the CAT exam in Delhi. 50% of the boys and 90% of the girls cleared the cut-off. If the total percentage of qualifying students is 60%, how many girls appeared in the examination?

Options: 100, 120, 150, 200

By alligation, ratio of Boys : Girls = $(90 - 60) : (60 - 50) = 30 : 10 = 3 : 1$

Total students = 800, divided in ratio 3:1 $\rightarrow$ total parts = 4

Girls = $(1/4) \times 800 = 200$

Answer: 200

✦ **Trick / Shortcut**

ALLIGATION RULE (very important): if two groups with percentages p_1 and p_2 mix to give overall percentage p , then $\text{Quantity}_1 : \text{Quantity}_2 = (p_2 - p) : (p - p_1)$.

Draw the alligation cross: put p_1 and p_2 at the ends, p in the middle, and cross-subtract diagonally to get the ratio instantly — no equations needed.

Q19. If 10 kg of sugar costing Rs.15/kg and 20 kg of salt costing Rs.10/kg are mixed, find the average cost of the mixture in Rs. per kg.

Options: 11.67, 12.33, 12.67, 11.33

Total cost = $(10 \times 15) + (20 \times 10) = 150 + 200 = 350$

Total quantity = $10 + 20 = 30$ kg

Average cost per kg = $350 \div 30 = 11.67$

Answer: 11.67

✦ **Trick / Shortcut**

Average (mean) price of a mixture = Total Cost $\div$ Total Quantity — same weighted-average idea as Q10.

Sanity check with alligation: since salt (20kg) is double sugar (10kg), the average should lie closer to Rs.10 than Rs.15 — 11.67 fits that check.

Q20. A tank contains 500 litres of wine. 50 litres of wine is taken out and replaced by water. The process is repeated again. Find the proportion of water and wine in the resulting mixture.

Options: 1:4, 41:50, 19:81, 81:19

This is a 'successive replacement / dilution' situation.

Formula: Final quantity of original liquid left = Initial $\times (1 - x/V)^n$, where x = volume removed each time, V = total volume, n = number of operations.

Wine left after 2 operations = $500 \times (1 - 50/500)^2 = 500 \times (0.9)^2 = 500 \times 0.81 = 405$ litres

Wine fraction = $405/500 = 81\%$, so Water fraction = 19%

Proportion of Water : Wine = 19 : 81

Answer: 19 : 81

✦ **Trick / Shortcut**

Memorise the replacement formula: Final = Initial $\times (1 - x/V)^n$. It applies to milk-water, wine-water, or any repeated dilution problem.

Here $(1 - 50/500) = 0.9 = 9/10$, squared = $81/100$ — wine remains 81%, water becomes 19% directly, without tracking litres removed each round.

Q21. A man purchased a table and a chair for Rs.2000. He sold the table at a profit of 20% and the chair at a profit of 40%. In this way, his total profit was 25%. Find the cost price (in Rs.) of the table.

Options: 1500, 900, 1000, 800

By alligation on the profit percentages: Table% : Chair% ratio = $(40 - 25) : (25 - 20) = 15 : 5 = 3 : 1$

So Cost of table : Cost of chair = 3 : 1 (the item closer to the overall average gets the larger share... actually the item whose % is FARTHER from the mean gets the larger ratio part on its own side)

Total cost = 2000, split in ratio 3:1 $\rightarrow$ total parts = 4

Cost of table = $(3/4) \times 2000 = 1500$

Answer: 1500

✦ **Trick / Shortcut**

Alligation works for profit percentages too, not just prices/concentrations — treat the two profit% like two 'ingredient strengths' mixing to the overall profit%.

Cross-subtract: table's ratio part = (chair% - mean%), chair's ratio part = (mean% - table%).

Q22. In what ratio should two qualities of tea having the rates of Rs.40 per kg and Rs.30 per kg be mixed in order to get a mixture that would have a rate of Rs.35 per kg?

Options: 1:2, 1:1, 1:3, 3:1

By alligation: Ratio = (30 – 35 taken positively) : (35 – 40 taken positively) → use (Cheaper's distance) : (Dearer's distance)

Ratio of Rs.40 tea : Rs.30 tea = (35 – 30) : (40 – 35) = 5 : 5 = 1 : 1

Answer: 1 : 1

✦ **Trick / Shortcut**

Since Rs.35 is exactly the midpoint of Rs.40 and Rs.30, the mixing ratio must be 1:1 — you can spot this instantly without any calculation whenever the required price is the exact average of the two given prices.

Q23. Two vessels contain a mixture of spirit and water. In the first vessel the ratio of spirit to water is 8:3 and in the second vessel the ratio is 5:1. A 35 litre cask is filled from these vessels so as to contain spirit and water in the ratio of 4:1. How many litres are taken from the first vessel?

Options: 11 litres, 22 litres, 16.5 litres, 17.5 litres

Spirit fraction in vessel 1 = 8/11. Spirit fraction in vessel 2 = 5/6. Required spirit fraction in cask = 4/5.

Let x litres be taken from vessel 1, so (35 – x) litres from vessel 2.

Spirit balance: $x \cdot (8/11) + (35-x) \cdot (5/6) = 35 \cdot (4/5) = 28$

Multiplying throughout by 66 (LCM of 11 and 6): $48x + 55(35 - x) = 1848$

$48x + 1925 - 55x = 1848 \rightarrow -7x = -77 \rightarrow x = 11$

Litres taken from the first vessel = 11 litres

Answer: 11 litres

✦ **Trick / Shortcut**

For mixtures of mixtures, always convert ratios to FRACTIONS of the required component (here spirit) before applying alligation/equations — it avoids errors.

Alligation shortcut: ratio of quantities taken = $(5/6 - 4/5) : (4/5 - 8/11)$; simplifying gives the same 11:24 split of 35 litres → 11 litres from vessel 1.

Q24. Two buckets of equal capacity are full of a mixture of milk and water. In the first, the ratio of milk to water is 1:7 and in the second it is 3:8. Now both mixtures are mixed in a bigger container. What is the resulting ratio of milk to water?

Options: 35:141, 42:49, 43:41, 41:53

Take a common capacity for both buckets, say 88 litres (LCM of 1+7=8 and 3+8=11).

Bucket 1: Milk = $88 \times 1/8 = 11$ L, Water = $88 \times 7/8 = 77$ L

Bucket 2: Milk = $88 \times 3/11 = 24$ L, Water = $88 \times 8/11 = 64$ L

Total Milk = 11 + 24 = 35 L; Total Water = 77 + 64 = 141 L

Resulting ratio Milk : Water = 35 : 141

Answer: 35 : 141

✦ **Trick / Shortcut**

When two ratios must be combined, pick a common total = LCM of (sum of parts of ratio 1, sum of parts of ratio 2) — this makes both buckets whole numbers and avoids fractions.

Here LCM(8, 11) = 88, chosen directly as the equal capacity of each bucket.

Q25. Sharman decides to travel 100 kilometres in 8 hours, partly by foot and partly on a bicycle, his speed on foot being 10 km/h and that on bicycle being 20 km/h. What distance would he travel on foot?

Options: 20 km, 30 km, 50 km, 60 km

Let time on foot = t_1 hours, time on bicycle = t_2 hours; $t_1 + t_2 = 8$

Distance equation: $10t_1 + 20t_2 = 100$

Substitute $t_1 = 8 - t_2$: $10(8 - t_2) + 20t_2 = 100 \rightarrow 80 + 10t_2 = 100 \rightarrow t_2 = 2$, so $t_1 = 6$

Distance on foot = speed $\times$ time = $10 \times 6 = 60$ km

Answer: 60 km

⚡ **Trick / Shortcut**

This is 'alligation on average speed': average speed overall = $100/8 = 12.5$ km/h.

Ratio of time on foot : time on bicycle = $(20 - 12.5) : (12.5 - 10) = 7.5 : 2.5 = 3 : 1$, i.e. of the 8 hours, foot time = $(3/4) \times 8 = 6$ h $\rightarrow$ distance = 60 km. Alligation gives the same answer instantly.